Medication Management for Older Adults: Literature Review

Key Takeaways and What This Means for My Research

After going through all 25 papers, a few things stand out that directly shape the direction of my robot-assisted medication management project.

The core problem is real and well-documented. Older adults managing medications at home deal with polypharmacy, cognitive decline, and physical limitations that lead to frequent errors, especially missed doses. About 18% of home-dwelling older adults miss at least two doses per week, and adherence rates range wildly depending on how much support someone has (from as low as 25% to as high as 97% with pharmacist involvement).

Technology helps, but the evidence is messy. Medication management technologies (pillboxes, apps, smart dispensers, robots) do seem to improve adherence overall, but the certainty of evidence is rated "very low" by formal quality standards. Effect sizes are all over the place, and most studies have small samples and short durations. So we can say these tools probably help, we just can't say exactly how much or for whom.

The biggest gap is exactly where this project sits. Very few studies look at using computer vision or AI to verify that someone has the right pills. Most existing tech focuses on reminding people to take medications, not on checking that they're taking the correct ones. Pill identification using deep learning and pill counting using vision models come up in only a couple of papers, and none of them test those capabilities in a real home-robot scenario. That's the gap my project can start to fill.

Simpler tools work fine for cognitively healthy people. Robots and AI matter most for those with cognitive decline. Studies consistently show that people with mild cognitive impairment benefit the most from automated, reminder-heavy systems, while cognitively intact older adults do fine with basic organizers. So the robot-assisted approach I'm exploring is most justified for the people who need it the most.

Technology dependence is a real concern. One study showed adherence jumped to 96% during the intervention but dropped back to 77% once the technology was removed. These tools support people but don't teach them new habits. That's not necessarily a problem for a robot that's always present, but it's worth keeping in mind.

Trust is task-specific. Older adults actually prefer robot help for medication reminders, but they want human help for deciding what medication to take. This fits well with what my project is doing. The robot isn't making medical decisions. It's verifying pills and reminding people, which is exactly where older adults are most open to automation.

Design for this population means big fonts, multiple reminder types, and caregiver integration.

Every successful system in the literature uses large text, high contrast, audio and visual cues, and some way to notify a caregiver when something goes wrong.

The Problem: Why Medication Management Is Hard for Older Adults

Polypharmacy is the challenge that comes up the most across the literature. When older adults take multiple medications on different schedules, keeping track of what to take and when gets complicated fast [1-3, 6, 16, 20, 23-25]. On top of that, cognitive decline makes things harder, especially problems with prospective memory, which is basically the ability to remember to do something in the future, like taking a pill at 2pm [10]. Physical limitations like reduced dexterity and impaired vision also make it harder to handle pills, read labels, and open packaging [3, 7, 9, 12].

Caregiver support is inconsistent. Family caregivers often don't know enough about the medications their loved ones are taking [2], and caregiver burden limits how much help they can realistically provide [1]. Communication gaps between different levels of the healthcare system add to the problem [8].

Common Medication Errors

The most frequent error is simply missing a dose. Omission is the most common mistake among older adults [11], with 18% of home-dwelling older adults missing at least 2 doses per week [4] and 23% reporting difficulty remembering to take their medications [4].

Other error types include wrong dosage, wrong timing, and drug-drug interactions [8, 9]. Some older adults accumulate medications they should have stopped taking, confuse medications that look similar, or even give their medications to others [10]. Wrong dose or medication errors accounted for 5% of hospital admissions in one study [8].

Adherence rates vary a lot depending on the population and how much support they have. Young-old adults (roughly 65-75) showed 94% adherence in one study, compared to 85% in old-old adults (75+) [11]. Across a systematic review, about one-third of patients were non-adherent [3]. The range goes from 25% adherence in institutionalized patients with minimal support to 98% in pharmacist-led programs [5]. Non-adherence is estimated to cost $100 billion annually in the U.S. [19].

What Technologies Have Been Tried

Electronic pillboxes and organizers are the most studied category. One review found 17 different electronic pillbox systems [6], from simple over-the-counter organizers to devices with multiple alarm types and locking mechanisms. These are the most mature products but often aren't designed well for older adults [6].

Smart medication dispensing systems go further with real-time tracking, cloud-based data syncing, and automated dispensing. A scoping review found 114 smart adherence products on the market or in development, with about 70% already being sold commercially [7].

Robotic systems are the most relevant category to my project. The Evondos E300 is the best-studied one. It's a robotic dispenser classified as a Class I Medical Device in the EU. It distributes pre-packaged medicine sachets on schedule, monitors whether doses are retrieved, and can alert healthcare providers if a dose is missed [4].

Mobile apps like the ALICE app let patients see images of their medications, set personalized alerts, and get drug interaction warnings [8]. Some apps also use machine learning for adherence prediction [8].

Ambient display systems take a different approach. Instead of explicit reminders, they use abstract cues like pictures, sounds, and movement to trigger the memory of taking medication [10]. The idea comes from research on distributed cognition, where environmental features support memory without requiring conscious effort [21].

AI-based systems are emerging but still early. These include personalized reminders through reinforcement learning, pill identification using deep neural networks, and drug interaction detection using natural language processing [9]. Machine learning models have also been used to predict which patients are likely to become non-adherent, with accuracy (AUC) up to 0.935 [5].

How Well Do These Technologies Work

The short answer: they generally help, but the evidence quality is low and the results are all over the place.

The clearest positive result comes from the ambient display study (Zarate-Bravo et al.), where adherence went from 81% at baseline to 96% during the intervention in a group of older adults with mild cognitive impairment [10]. But when the technology was removed, adherence dropped back to 77%, which is actually below baseline. That raises questions about whether these gains last [10].

The Evondos robotic dispenser achieved 98.7% on-time dose delivery in home settings [4], and delivery failures dropped 63% compared to earlier nursing home testing [4]. The ALICE app showed statistically significant improvements in adherence scores [8], and 88% of users reported feeling more independent in managing their medications [8].

At the meta-analysis level, Mehany et al. pooled 10 randomized trials and found an overall positive effect on adherence, but with huge heterogeneity (I-squared=99%) and very low certainty of evidence by GRADE criteria [5]. Basically, the results are scattered, and the studies themselves have quality issues. Only 2 of the 10 trials were rated low risk of bias [5].

One pattern worth noting: passive supports (automated reminders, external cues) seem to work better than educational interventions for this population. Memory aids and cues improved adherence in 4 out of 5 studies, while educational interventions only worked in 12 out of 28 [22]. This tracks with what we know about older adults doing better when they can rely on preserved automatic memory processes rather than having to learn new things [10].

There's also a technology readiness paradox worth mentioning. The electronic pillboxes with the most clinical evidence have high technology readiness (mean TRL 8.9), but weren't designed for older adults specifically. Newer products designed with older adults in mind are stuck at lower readiness levels (mean TRL 5.6) and don't have enough clinical validation yet [6].

Why People Don't Use These Technologies

Usability is the biggest barrier. Pillboxes often have small compartments that are hard to use with reduced dexterity [6]. Instructions are unclear, labeling is poor, and many devices aren't built for older adults managing complex regimens [12]. Some robotic systems also can't handle liquid medications [1].

Trust and acceptance come up a lot. Older adults worry about privacy [1] and don't love systems that reduce human contact [1]. But trust depends on the task. People are more comfortable with robots handling reminders than with robots making medication decisions [18]. It also depends on whether someone feels capable and whether they see the robot as reliable [18].

Cost is a real barrier, especially for low-income older adults. A lot of electronic pillboxes are too expensive [6], and smart systems often have subscription charges on top of the device cost [7].

Digital literacy is a challenge, but not always in the way you'd expect. While many apps are designed for more tech-savvy users [8], the ALICE study found that older adults with no prior tech experience actually did better with the app than those who had experience [8]. One possible explanation is that tech-naive users paid more attention to the full system instead of skipping over features they assumed they already understood.

Where AI and Robotics Currently Stand

Robotic medication systems have reached real-world deployment in some cases. The Evondos

E300 is the most concrete example. It handles pre-packaged sachets, provides spoken and visual reminders, monitors retrieval in real-time, and alerts providers when doses are missed [4]. It works, but it's limited to pre-packaged medications and needed a nursing home pilot before moving to home deployment [4].

AI applications cover a few different levels. At the simplest, reinforcement learning personalizes when and how reminders are delivered based on how individual users respond [9]. For verification, deep neural networks combined with optical character recognition can identify medication containers [9]. Natural language processing can detect potential drug-drug interactions [9]. More recent work combines large language models with smartphone apps for broader medication management, including meal planning and activity suggestions [13].

Computer vision for pill identification is the area most relevant to my project, and it's also the least developed. The ALICE app uses image recognition to show patients pictures of their medications [8], and some systems use QR codes for real-time medication information [12]. But I didn't find any studies that test vision-based pill counting or color-based pill verification in a home robot context. That's the specific gap my project is trying to address.

What Good Design Looks Like for This Population

A few design principles come up consistently across the literature and should inform how I build my robot system.

Accessibility first. Large fonts, high contrast, simple tap interactions, and screens that are 7 inches or larger [8]. Colors and bold labeling that are easy to see [12]. Audio and visual cues used together so the system works for people with different sensory abilities [3, 6].

Respect existing routines. Ethnographic work shows that older adults build their own spatial and temporal systems for managing medications. They place pills near the coffee maker, link doses to meal times, and use physical locations as memory triggers [21]. Technology that works well supports these routines instead of replacing them.

Use multiple reminder types. Systems that combine spoken reminders, sound signals, light signals, and written instructions do better than systems that only use one type [4, 6]. One study found that audio feedback described as the "voice of a friend" was more acceptable than generic alarm tones [6].

Build in caregiver integration. The systems that work best have a way to notify family members or healthcare providers when something goes wrong, whether through app notifications [8], QR-code-based information sharing [12], or real-time monitoring dashboards [13].

Include fail-safe mechanisms. Given the vulnerable population, systems need multiple safety layers: reminders to prevent missed doses, verification to prevent wrong-drug errors, interaction checking, and escalation to human caregivers when the system isn't sure [9]. The Evondos device stores unused doses and sets off malfunction alarms [4].

Personalize where possible. AI systems that adapt their timing and communication style to individual users work better than fixed-schedule approaches [9]. But if personalization is based only on existing routines, it can accidentally reinforce bad medication habits, so clinical oversight is still needed [10].

What's Still Missing (and Where My Research Fits)

Based on this review, a few main gaps stand out.

No studies test vision-based pill counting or verification in a real robot scenario. AI pill identification exists in lab settings [9], but nobody has tested whether a robot can look at pills, count them, identify them by color, and flag uncertainty in a real home-care context.
There's very little work on uncertainty and error detection. Most systems either work or they don't. The connection between physical pill arrangement and vision model accuracy hasn't been looked at. There's very little on how robot form factor affects trust and engagement. The literature shows that older adults have specific preferences about what tasks they want robots to do, but there's almost no data on how a particular robot like the Reachy Mini compares to a screen-based or voice-based system in terms of trust and comfort.

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